

Input Order Is an Uncontrolled Variable in Lossless Point Cloud Compression Benchmarks

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Abstract

Lossless point cloud codecs are compared at equal bitrate, but they do not all encode the same object: some reconstruct the multiset of points, others the sequence. The arrangement of the input is not part of any published test conditions, so a reported bitrate is not determined by them. We make the arrangement an explicit factor — eight order conditions on a fixed point set, four lossless coders, 26 blocks from 12 files, 204 rows — and give the definitions needed to repeat the measurement.

The effect dwarfs the differences the benchmarks are built to resolve. At the file’s own order the four coders lie within a median of 13% of each other, and no coder wins everywhere: an octree is shortest on 14 of 26 blocks, a sequential context coder on 6, a scan-model coder on 6. Permuting the same points uniformly costs the sequential coders a median of +121.9% and +130.9%, an order of magnitude more than the spread between coders. The octree agrees with the original order to three digits in all 204 rows.

Measuring this needs two null controls. The seed-to-seed difference between two random permutations has a median of 0.015 percentage points, but reversing the input — which preserves every local neighbourhood — moves one coder by -1.28 to $+1.59$. The larger is the resolution of the experiment. Against it, the scan-model coder is indistinguishable from a benign rearrangement on every block whose sort key is at most 45% duplicated except one; it sits 4.4 times the floor at 82.4% duplication and 320 to 495 times at full degeneracy, where it collapses onto a plain geometry coder. A controlled sweep that manufactures key duplication while holding the input order fixed reproduces the collapse at full degeneracy and returns the cost to zero when uniqueness is restored at the same quantisation step.

1 Introduction

A lossless point cloud codec is judged by bits per point (bpp). The comparison assumes that the coders under test solve the same problem. They do not.

An octree coder such as G-PCC reconstructs the multiset of occupied positions: the order in which the points arrived is discarded. A sequential coder such as LASzip reconstructs the sequence, so that point i of the output is point i of the input. For $n = 10^6$ the difference is

$$\frac{\log_2 n!}{n} \approx \log_2 n - \log_2 e = 18.489 \quad (1)$$

bpp of information that one carries and the other does not, comparable to the entire geometry budget of an airborne LiDAR tile.

What is missing is a measurement. The common test conditions used to compare lossless point cloud coders fix the sequences, the quantisation and the attribute set, but not the arrangement of the points within a file. We treat the arrangement as an experimental factor, give the definitions needed to reproduce the measurement, and report both the absolute code lengths and their response to order.

1.1 Contributions

The gap in the literature is narrow and we state it narrowly: the measurement. Every component of the order-invariance property is prior art (Section 2).

1. A protocol that makes input order an explicit factor, with two null controls and a stated resolution, defined in Section 3.
2. The measurement it produces: four coders, 26 blocks, eight conditions, 204 rows, with absolute code lengths alongside the relative response (Section 6).
3. A quantification of how a key-sorted coder loses order invariance as its key degenerates, with a controlled sweep that manipulates the key alone (Section 7).

2 Related work

Order-invariant geometry coding is standard. The octree of G-PCC and the kd-tree of Draco both code a multiset, so their geometry cost does not depend on the input arrangement. TMC13 additionally offers `--sortInputByAzimuth` and `--predGeomSort`, which re-derive an acquisition-like order internally, and a `canonical_point_order_flag`. Re-deriving a scan order from the data is prior art, including in recent learned coders [4].

Transmitting the order as a separate channel is disclosed. US10796457B2 (Van Beek, Intel, priority 2018) states that if point positions are coded losslessly the reconstructed X, Y, Z are identical whatever the coding order, and that the choice of coding order may be transmitted so that the original order can be

reconstructed. We have not analysed the independent claims and make no statement about claim scope.

The log $n!$ framework is established in the opposite direction. Shuffle coding [2] and Random Order Coding [3] spend effort to *save* the bits of (1) for an unordered set. We measure what it costs to keep them.

LASzip [1] is the representative order-dependent coder: it does not re-sort, and its predictor follows the scan line. Re-sorting is left to external tools, which destroy the input order.

Our own earlier work [5] reports one instance of the effect (one file, 2M points, one direction) and claims order invariance for the attribute columns of a coder whose coding order is derived from the geometry. The present paper measures the geometry columns, across coders and conditions, and identifies the condition under which the invariance fails.

3 Formulation

3.1 Setting

A point cloud is a finite *sequence* $x = (x_1, \dots, x_n)$ with $x_i \in \mathbb{Z}^3 \times \mathcal{A}$, where the integer triple is the quantised coordinate and $\mathcal{A}$ holds the attributes. Write $x_i^g \in \mathbb{Z}^3$ for the coordinate part.

A codec is a pair (E, D) . It is *sequence-lossless* if $D(E(x)) = x$ and *multiset-lossless* if $D(E(x))$ equals x as a multiset. Its code length is reported in bits per *point* (bpp throughout this paper; not bits per pixel),

$$L(x) = \frac{8|E(x)|}{n} \quad [\text{bpp}], \quad (2)$$

where $|E(x)|$ is the size of the coded file in bytes and n is the number of points *presented to the encoder*, counting coincident points separately. The same n divides all four coders, and none of them merges coincident points: the blocks contain up to 9,850 of them and every round trip reproduces them.

For $\pi \in S_n$ let $x \circ \pi = (x_{\pi(1)}, \dots, x_{\pi(n)})$. The quantity we measure is the relative change of code length under a rearrangement,

$$\Delta(\pi) = 100 \left(\frac{L(x \circ \pi)}{L(x \circ \text{id})} - 1 \right) \quad [\%], \quad (3)$$

with the file's own order as the baseline. A multiset-lossless codec is not *required* by its definition to have $\Delta \equiv 0$; whether it does is an empirical question about the implementation, and we measure it rather than assume it.

3.2 Order conditions

Eight conditions, all on the same point set.

Identity. $\pi = \text{id}$, the order in the source file.

Reverse. $\pi(i) = n + 1 - i$.

Morton. π sorts the points by the Morton code of their coordinates. Each axis is first normalised to its own range, which matters because raw coordinates can

exceed the width of the interleaved word:

$$b_{i,c} = x_{i,c}^g - \min_j x_{j,c}^g, \quad r_c = \max_j \left(\max_j b_{j,c}, 1 \right), \quad (4)$$

$$a_{i,c} = \left\lfloor b_{i,c} (2^{21} - 1) / r_c \right\rfloor, \quad (5)$$

and $m(i) = \sum_{c=0}^2 \sum_{k=0}^{20} ((a_{i,c} \gg k) \& 1) \ll (3k + c)$ is the interleaving. The sort is stable.

Window w . Independent uniform permutations inside consecutive blocks of w indices: π restricted to $\{jw + 1, \dots, \min((j+1)w, n)\}$ is uniform on that block, independently for each j . It interpolates between the identity ($w = 1$) and a uniform permutation ($w = n$). We use $w \in \{10^2, 10^3, 10^4\}$ and apply a condition only when $10w \leq n$, so that a window is never a substantial fraction of the block.

Uniform. $\pi \sim \text{Uniform}(S_n)$, drawn twice with independent seeds.

3.3 Resolution

A difference in Δ is only meaningful above the variation the apparatus produces on its own. We use two controls.

The *seed control* is the difference between the two uniform draws,

$$\sigma = \max_{\text{blocks}} |\Delta(\pi_1) - \Delta(\pi_2)|, \quad (6)$$

which measures variation within one treatment, with the baseline cancelling.

The *reverse control* is

$$\rho_{\text{rev}} = \max_{\text{blocks}} |\Delta(\text{rev})|. \quad (7)$$

Reversal preserves every local neighbourhood, so a direction-symmetric sequential predictor should be unaffected; the deviation measures what a benign rearrangement costs. The *effective floor* is

$$\varphi = \max(\sigma, \rho_{\text{rev}}), \quad (8)$$

computed per coder. We call an observed $|\Delta|$ resolved only if it exceeds φ .

3.4 Key duplication

One of the coders sorts internally by a key $k_i = (s_i, t_i)$, the flight line identifier and the timestamp. Its degeneracy is

$$\rho_{\text{key}} = 1 - \frac{|\{k_1, \dots, k_n\}|}{n} \in [0, 1), \quad (9)$$

the fraction of points whose key is shared with an earlier one.

Proposition 1 *If $\rho_{\text{key}} = 0$ and the coder's internal sort is a stable sort on k , then $L(x \circ \pi) = L(x)$ for every $\pi \in S_n$.*

With $\rho_{\text{key}} = 0$ the key is injective, so it induces a total order; a stable sort returns that order whatever the input arrangement, and the encoder sees the same sequence. Stability is irrelevant here but becomes the whole question when $\rho_{\text{key}} > 0$, because ties are then broken by input position. Sections 6 and 7 measure what happens in that regime.

4 The four coders

All four are measured on geometry alone, so that the levels are comparable.

G-PCC. TMC13 `release-v23.0-rc2`, lossless octree, geometry only, driven through a PLY file with `float32` coordinates, which bounds the integer range the harness can carry. Multiset-lossless.

LAZ. LASzip through a LAS 1.4 point record of type 6 containing only X, Y, Z . The figure is the whole file divided by n , so it includes a header and a chunk table that do not depend on the order; on the smallest block this fixed part is about 0.3 bpp. Sequence-lossless.

v3, our sequential context coder. Points are coded in input order; for each point the three axes are coded in turn. Let $d_{i,c} = x_{i,c}^g - x_{i-1,c}^g$ and let the predictor be the running median of the last three differences,

$$\hat{x}_{i,c} = x_{i-1,c}^g + \text{med}(d_{i-1,c}, d_{i-2,c}, d_{i-3,c}). \quad (10)$$

The residual is mapped to a non-negative integer by $\zeta(v) = 2v$ for $v \geq 0$ and $-2v - 1$ otherwise, and $z_{i,c} = \zeta(x_{i,c}^g - \hat{x}_{i,c})$ is written by an adaptive range coder with a context selected by magnitude. With $\kappa(z) = \min(\lceil \log_2(z+1) \rceil, 23)$ giving 24 buckets per axis, the context index is

$$\gamma_{i,c} = \begin{cases} \kappa(z_{i-1,0}) & c = 0, \\ \kappa(z_{i,0}) & c = 1, \\ \lfloor (\kappa(z_{i,0}) + \kappa(z_{i,1}))/2 \rfloor & c = 2. \end{cases} \quad (11)$$

Axis 0 takes its context from the previous point; axes 1 and 2 take theirs from the axes already coded at the same point. This cross-axis coupling is what distinguishes v3 from a plain first-difference coder.

scan, our scan-model coder. It sorts the points by k_i with a stable sort, splits the sorted sequence into sweeps wherever the timestamp gap exceeds 20 times the median firing interval, and fits a model per sweep. Within a sweep, the horizontal coordinates are rotated into the sweep plane by three integer shears, each a bijection of $\mathbb{Z}^2$, so the rotation is exactly invertible without floating point. Writing s for the in-plane coordinate, z for height, S_z for the sensor height, θ_0 for the scan angle at the start of the sweep and ω for the angular step per shot, the prediction for the j -th shot of the sweep is

$$\hat{s}_j = s_0 + \lfloor (S_z - z_j) \tan(\theta_0 + \omega j) \rfloor, \quad (12)$$

with the tangent read from a fixed-point table built by an integer CORDIC, so the decoder never evaluates a transcendental function. The five parameters $(s_0, S_z, \theta_0, \omega)$ and the shear constants are fitted per sweep by Levenberg–Marquardt and written as side information; a sweep whose fit does not pay for itself falls back to (10). Residuals are coded as in (11). Sequence-lossless.

4.1 What is compared

Table 1 separates what the decoder must reconstruct from whether the code length depends on the arrangement. A coder in the top row is not comparable at equal bitrate with one in the bottom row, because the bottom row also transmits the permutation, worth (1).

Table 1: Lossless coders divide on two independent axes. A comparison at equal bitrate is meaningful only within a row.

	Cost depends on order	Cost independent
Multiset	—	G-PCC (single slice)
Sequence	LAZ, v3	scan, conditionally

Table 2: Geometry code length at the file’s own order, in bpp. The three left columns are over the 21 blocks of 10^6 points; the two right columns count the blocks on which each coder is shortest, over those 21 and over all 26.

	median	min	max	best/21	best/26
G-PCC	24.451	6.866	27.019	10	14
LAZ	23.011	7.764	36.615	0	0
v3	22.860	7.531	37.025	6	6
scan	23.338	7.410	36.920	5	6

5 Experimental setup

Twelve files, all in the LAS/LAZ family: airborne LiDAR tiles (AHN3, AHN4 $\times 2$, AHN5, Autzen $\times 2$, Red Rocks, a survey block carrying extra bytes) and five small test files, some synthetic. Seven large files contribute three disjoint blocks of 10^6 points each, below the default `sliceMaxPoints` of TMC13 so that G-PCC codes a single slice; five small files are used whole (10,683 to 110,000 points). This gives 26 blocks, and $26 \times 8 - 4 = 204$ rows once the four window conditions that the small blocks cannot support are removed. The four coders are columns.

Every row is decoded and compared. For our own coder the candidate is pinned, so the configuration reported is the configuration verified, and the fallback that would substitute an embedded container is disabled. “Verified” means equality of all columns in order for LAZ and for our container, and equality of the multiset of coordinates after a lexicographic sort for G-PCC. All 204 rows pass.

6 Results

6.1 Absolute code length

Table 2 gives the levels the benchmark is trying to resolve. Over all 26 blocks, the median difference from G-PCC on the same block is +12.8% for LAZ, +8.7% for v3 and +7.0% for scan, and each sequential coder is shorter than G-PCC on some blocks and longer on others (6, 10 and 9 of 26 respectively; LAZ is never the shortest of the four because v3 differs from it only by the cross-axis context of (11), which is never harmful on this data). The extremes are wide in both directions: the scan coder is 30.9% shorter than G-PCC on one airborne block (17.098 against 24.755 bpp) and 41.8% longer on a small vegetation sample (29.485 against 20.791). No coder dominates.

For cost: at 10^6 points the medians are 1.9 s to encode and 1.6 s to decode with a peak of 821 MB for G-PCC, under 0.05 s each way for LAZ, and 10.4 s / 0.9 s with a peak of 1.6 GB for our container. The LAZ peak cannot

Table 3: Code length relative to the file’s own order, per cent, by (3). Median over 26 blocks; the window 10^4 row is over the 22 blocks large enough for that condition. The last row gives the interquartile range of the uniform condition.

Condition	G-PCC	LAZ	v3	scan
reverse	+0.0	−0.0	−0.0	−0.1
Morton	+0.0	+16.1	+23.0	−0.0
window 10^2	+0.0	+65.5	+70.7	+0.3
window 10^3	+0.0	+80.8	+86.0	+0.3
window 10^4	+0.0	+89.1	+95.6	+0.2
uniform 1	+0.0	+121.9	+130.9	+0.2
uniform 2	+0.0	+121.9	+130.9	+0.2
IQR, uniform	—	[89, 140]	[102, 153]	[0.0, 1.5]

Table 4: The two null controls of Section 3.3, in percentage points, and the effective floor (8).

	σ med.	σ	rev. min	rev. max	φ
G-PCC	0.000	0.000	+0.000	+0.000	0.000
LAZ	0.009	0.664	−0.385	+0.588	0.664
v3	0.015	0.261	−1.099	+0.070	1.099
scan	0.015	0.305	−1.282	+1.591	1.591

be separated because its implementation runs inside the measuring process.

6.2 Order sensitivity

Table 3 is the main result. A uniform permutation of the same points costs LASzip a median of 121.9% and v3 130.9% of their original-order geometry budget, with extremes of +36.0% and +828.6% over the 26 blocks. Set against Table 2, where the four coders lie within a median of 13% of one another, the arrangement of the input is an order of magnitude more important than the choice of coder.

G-PCC agrees with the original order to three significant digits in every one of the 204 rows. At 10^6 points three digits is a resolution of 0.001 bpp, or 125 bytes, so the statement is “agrees to three digits”, not “exactly zero”.

6.3 Resolution

The two controls differ by a factor of five for the scan coder (Table 4). Nine of its 26 blocks move further under reversal than the largest seed-to-seed difference, so a floor built from the seed control alone would count as responses differences that a neighbourhood-preserving rearrangement also produces: 15 blocks exceed $\sigma = 0.305$, but only 6 exceed $\varphi = 1.591$. All results below are stated against φ .

6.4 Window width and Morton order

Code length increases monotonically in the window width for LAZ (22 of the 22 blocks that support all four conditions) and for v3 (22 of 22). For the scan coder it is monotone on 3 of 22, and those three are the blocks of the one fully degenerate file large enough to enter the count, where the response is the largest effect

Table 5: Cost of a uniform permutation by file, against key duplication (9). The representative block of a file is the one whose duplication is the median; the last column gives the range over the three blocks where a file has three.

File	ρ_{key}	LAZ	v3	scan	scan, range
red-rocks	1.000	+458	+513	+517	+510 to +522
simple1_4	1.000	+829	+642	+787	—
plane	0.824	+189	+209	+7.0	—
workshop	0.392	+133	+146	+1.8	+0.3 to +1.8
fullwave	0.337	+69	+86	+0.5	—
AHN3	0.240	+123	+134	+0.7	+0.4 to +1.3
AHN4 (b)	0.183	+136	+149	−1.1	−1.1 to +0.1
AHN5	0.155	+116	+131	+0.2	+0.1 to +1.2
autzen-trim	0.097	+119	+128	−0.4	—
AHN4 (a)	0.083	+124	+135	−0.2	−0.3 to −0.1
autzen-2023	0.073	+84	+87	+0.0	+0.0 to +0.1
vegetation	0.000	+56	+56	+0.0	—

in the study (+225 $\rightarrow$ +317 $\rightarrow$ +413 $\rightarrow$ +522%). The other 19 are flat at the level of the floor.

Sorting into Morton order raises the median cost by 16.1% for LAZ and 23.0% for v3, but the sign is not uniform: Morton order shortens the code on 6 of 26 blocks for LAZ, 5 of 26 for v3 and 14 of 26 for scan, with a range for v3 of −21.0% to +232.3%. “A space-filling curve is bad for sequential predictors” is a statement about a median, not a rule.

7 Order invariance and key degeneracy

Table 5 measures the regime the proposition does not cover. Against the floor $\varphi = 1.591$:

- $\rho_{\text{key}} \leq 0.45$ (21 blocks): unresolved, with one exception at 1.1 φ ;
- $\rho_{\text{key}} = 0.824$ (1 file): 4.4 φ , and saturating across the window conditions (+6.86, +7.06, +7.01%);
- $\rho_{\text{key}} = 1$ (4 blocks): 320 to 495 φ , and graded across the window conditions.

At full degeneracy the coder loses the property outright: on red-rocks the scan coder (+517%) lands on the plain geometry coder (+513%), having collapsed into one. The saturating response at 0.824 and the graded response at 1 are not the same mechanism at two strengths.

The rank correlation between ρ_{key} and the cost of a permutation is +0.697 ($p = 0.025$) over the ten non-degenerate files for the scan coder. There are nine such tests in this paper, giving a Bonferroni bound of 0.23, and removing any one of four files from the ten destroys significance. It is reported for completeness and nothing rests on it.

7.1 A controlled sweep

Table 5 is observational: key duplication is confounded with everything else that distinguishes one survey from another. To break the confound we hold the input order

Table 6: Input order held at the identity; key duplication manufactured by quantising the timestamp to a target rate. Cost of a uniform permutation and of the reverse control at the same quantisation step, per cent.

ρ_{key}	vegetation		AHN4	
	uniform	reverse	uniform	reverse
0.25	-1.6	-4.6	+4.0	+4.5
0.50	-0.7	-4.8	+5.9	+6.2
0.75	+6.0	-2.0	+14.6	+13.9
1.00	+46.2	+0.1	+125.9	-0.1
dither at $\rho_{\text{key}} = 1$: +0.0 and +0.0				

at the identity and manufacture duplication by quantising the timestamp, with a dither condition that restores uniqueness at the same quantisation step as a control.

The $\rho_{\text{key}} = 1$ row is decisive. Manufacturing a fully degenerate key raises the cost of a permutation from nothing to +46.2 and +125.9%, and dithering at the same step returns it to zero. The effect is caused by degeneracy of the key, not by the loss of timestamp resolution that produced it. Note that the controlled effect is far smaller than the +509.7 to +787.4% observed on the two files that arrive with degenerate keys; those files differ in other ways as well, so the observational figure is not the effect of degeneracy alone.

The intermediate rows do not agree. On AHN4 the cost of a permutation tracks the reverse control to within a percentage point at 0.25, 0.50 and 0.75: no order-specific excess appears until the key is fully degenerate. On vegetation an excess grows (+3.0, +4.1, +8.0 percentage points over the control). The dither control does not fire at the intermediate levels on AHN4, because the quantisation step there is close to the floating-point spacing of the timestamps and there is no room to separate the groups again, so the intermediate part of the sweep has no working control on the file where the answer differs.

8 Implications for benchmarks

The order must be part of the test conditions.

A lossless bitrate reported without an order condition is not determined. A uniform permutation, the worst case, costs a sequential coder a median of +121.9%. The rearrangement that actually occurs in practice is milder and its sign is not fixed: sorting an acquisition-order file into a space-filling-curve order, which is what the standard tools do, costs a median of +16.1% and ranges from -21.4% to +304.1%.

Reordering the input is not a levelling operation. Sorting every input into a canonical order does not neutralise an advantage; it destroys the premise of the order-dependent coders, and the canonical order is not neutral either.

Compare within a row of Table 1. If a multiset coder is compared with a sequence coder, the 18.489 bpp of (1) must be accounted for on one side. The figure applies to the 21 blocks of 10^6 points and not to the five smaller ones, and it is the information content of

the permutation, not the cost a coder pays for disorder.

The multiset side is not exempt either. G-PCC is exactly order-invariant here because every block fits in one slice. Above `sliceMaxPoints` the assignment of points to slices follows input order, and real files exceed that threshold.

A minimum specification. An order clause needs to fix: (i) which arrangement is the baseline; (ii) whether coders may re-sort internally and whether the original order must be recoverable; (iii) how the permutation is accounted for when the two sides differ on (ii); (iv) a null control with a stated resolution; and (v) publication of the per-block values, not only summaries, since the medians in Table 3 conceal interquartile ranges of 50 percentage points.

9 Limitations

Sample size. Twelve files, of which four airborne tiles share a footprint and two share a site: eight independent sites, six among the non-degenerate. Detecting $\rho = 0.7$ at 80% power needs about 13. Nothing in this paper rests on the correlational evidence; the statements in Section 7 are made against the floor, block by block.

Confounding. In the observational data, key duplication is confounded with survey identity. The controlled sweep breaks the confound at full degeneracy only.

The gap in the key axis. The observational data contain nothing between $\rho_{\text{key}} = 0.45$ and 0.824, and one file above 0.45.

Accounting asymmetry. The scan coder’s figure covers X, Y, Z only, and those columns cannot be decoded without the timestamp and flight line that form the key. Including the side columns raises the cost of a uniform permutation to +137.9% in a separate pilot measurement whose per-candidate values were not round-trip verified; it is the only estimate available. The correct statement is conditional: given the side columns, the geometry cost of the scan coder is order-independent at the duplication rates where we can resolve it, $\rho_{\text{key}} \leq 0.45$.

Scope. LAS-family files only, and airborne LiDAR for every large block. No automotive or terrestrial laser scanning data, and no sequence from the MPEG common test conditions. Two of the four coders are our own and unpublished; the headline figure for the sequence row, LASzip at +121.9%, does not depend on them.

Novelty. Every component of the invariance property is prior art (Section 2). What is new is the measurement, the protocol, and the quantification of the degeneracy regime.

10 Conclusion

Input order moves the code length of sequential lossless point cloud coders by a median of +121.9% and +130.9% under a uniform permutation, an order of magnitude more than the differences between the

coders that benchmarks are built to resolve. It is absent from the test conditions used to compare them, and it interacts with coder internals in ways that are not monotone in sign. Making it explicit costs one axis in the experimental design and two null controls; leaving it implicit makes the reported numbers irreproducible.

References

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